Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	112.5 ÷ 100 × 152 oe	**1**	
1(a)(ii)	3 ÷ 8 × 152 + 152 + 171	**2**	**B1** for 3 ÷ 8 × 152
1(b)	8 : 9 : 3	**2**	**B1** for 152 : 171 : 57
1(c)	8	**2**	**M1** for correct attempt e.g. 12 + 14 + 16 ….
1(d)(i)	6 ÷ 4 × 7	**1**	
1(d)(ii)	13.50	**1**	
1(d)(iii)	3228	**3**	**FT** *their* **(d)(ii)** **M2** for 120 × *their* **(d)(ii)** + 136 × 10.5 + 30 × 6 or **B1** for *their* 1620 or 1428 or 180
1(d)(iv)	71.7 or 71.7[3.....]	**1**	**FT** *their* **(d)(iii)**
2(a)	Kite	**1**	
2(b)(i)	Translation $\begin{pmatrix} 4 \\ 9 \end{pmatrix}$	**2**	**B1** for each
2(b)(ii)	Reflection $x = 0.5$ oe	**2**	**B1** for each
2(b)(iii)	Rotation 90° clockwise oe [centre] (0, 0) oe	**3**	**B1** for each
2(c)(i)	(−5, −6)	**1**	
2(c)(ii)	Image at (−5, 0), (−2, 3), (7, 0),(−2, −3)	**2**	**B1** for correct size, wrong position or correct shape with incorrect scale factor
3(a)	Equilateral	**1**	
3(b)(i)	4.1 to 4.5	**1**	
3(b)(ii)	10.25 to 11.25	**2**	**M1** for 0.5 × 5 × *their* **(b)(i)**

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Question	Answer	Marks	Partial Marks
3(b)(iii)	61.5 to 67.5	**2**	**FT** *their* **(b)(ii)** **B1** for 6 seen
4(a)	4 7 4	**2**	**B1** for one correct
4(b)	Correct curve	**4**	**B3FT** for 6 or 7 points correct or **B2FT** for 4 or 5 points correct or **B1FT** for 2 or 3 points correct
4(c)	$x = 1$ oe	**1**	
4(d)	−1.9 to −1.7 and 3.7 to 3.9	**2**	**B1** for each
5(a)(i)	68	**1**	
5(a)(ii)	61, 67, 71, 73	**2**	**B1** for 3 correct and none incorrect or 4 correct and one incorrect
5(b)(i)	67	**1**	
5(b)(ii)	64	**1**	
5(b)(iii)	65	**1**	
5(b)(iv)	72	**1**	
5(c)	$\frac{1}{7}$	**1**	
5(d)	3.722	**1**	
5(e)	56	**2**	**B1** for $56k$ or lists of multiples of 8 and 14 (at least 3 of each)
5(f)(i)	12	**1**	
5(f)(ii)	17	**1**	
6(a)	Tangent	**1**	
6(b)(i)	22π final answer	**2**	**M1** for $2 \times 11 \times \pi$
6(b)(ii)	40	**2**	**B1** for angle $OBC = 40^\circ$ or angle $BOG = 140^\circ$
6(b)(iii)	7.68 or 7.679 to 7.680....	**2**	**FT** *their* **(b)(ii)** and **(b)(i)** **M1** for $\frac{\textit{their}\ \textbf{(b)(ii)}}{360} \times \textit{their}\ \textbf{(b)(i)}$
6(c)(i)	Angle [between] tangent [and] radius	**1**	